04_three_country_falsification

October 5, 2026

1 Three-country falsification: $\mathcal{S}_c \approx \mathcal{S}_G \oplus \mathcal{S}_{L,c}$

Run directory: `week5/runs/three_country_falsification_20261005_185246`

Full report + analysis figures (no recompute required).

- Report: `../runs/three_country_falsification_20261005_185246/REPORT.md`
- Config: `../runs/three_country_falsification_20261005_185246/config.json`
- Artifacts: `../runs/three_country_falsification_20261005_185246/artifacts/`

2 Three-country falsification — Global ⊕ Local task subspace

Run: `three_country_falsification_20261005_185246` **Hypothesis:** `S_c ≈ S_G ⊕ S_L,c`
Prior bases: `/home/kv133/Alpha-Earth-Embeddings-Research/week5/runs/global_local_subspace_2026100`
Seeds: [0, 1, 2, 3, 4] · **max_per_class:** 5000 · **r*:** 1

2.1 Success criteria

Overall: `partial_mixed`

#	Criterion	Result	Evidence
1	Significant shared task-subspace structure	FAIL	λ1=0.986 (null95=0.987), λ2=0.982 (null95=0.982)
2	S_G retains strong LOCO cross-country signal	PASS	LOCO global F1=0.657 vs lr64=0.510
3	$S_{L,c}$ adds within-country information	PASS	within SG⊕SL=0.733 > SG=0.710
4	Matched local beats mismatched	PASS	mean diagonal advantage=0.009; targets with CI>0: 2/3

2.2 Interpretation guide

- **all_four** → enough evidence to justify scaling globally

- **partial_1_2** → transferable common subspace, weak local structure
- **partial_no_country_specific** → global+residual helps, but residual is not country-specific
- **contradicts** → shared supervised geometry not supported

2.3 Representation vs OOD

- Descriptive S_G (all three countries): spectrum, within-country, swap structure
- LOCO $S_G^{(-c)}$: all cross-country transfer claims

2.4 Diagonal advantage by target

target	r	matched F1	mismatched F1	advantage	95% CI
USA	1	0.837	0.827	0.011	[0.010, 0.012]
JPN	1	0.719	0.719	0.000	[-0.000, 0.001]
IND	1	0.642	0.627	0.015	[0.011, 0.018]

2.5 Rank phenomenon (exploratory)

- Late-k (k≥10) mean ΔF1 within=+0.0005, ΔF1 OOD=-0.0043
- Early-k (2–5) mean ΔGlobalOverlap=+0.5722

2.6 Figures

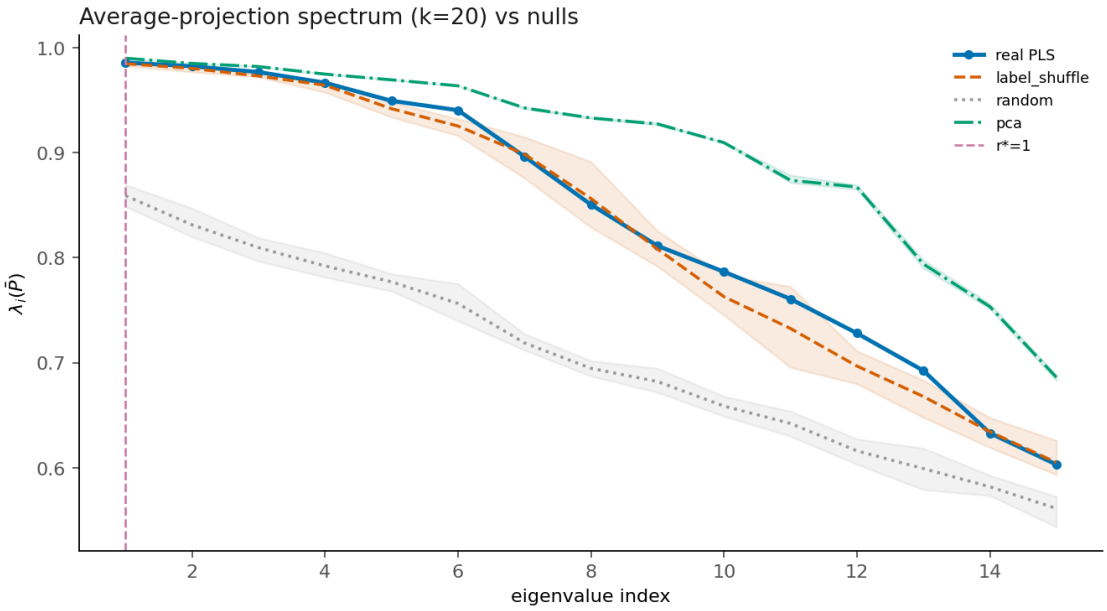
- `01_avgproj_spectrum_vs_nulls.png`
- `02_sg_sl_prediction_heatmap.png`
- `03_local_swap_matrix.png`
- `04_rank_f1_globaloverlap.png`
- `05_success_criteria_dashboard.png`

2.7 Reproducibility

- Config: `config.json`
- r*: `artifacts/r_star.json`
- Artifacts: `artifacts/*.csv`
- Notebook: `week5/notebooks/04_three_country_falsification.ipynb`

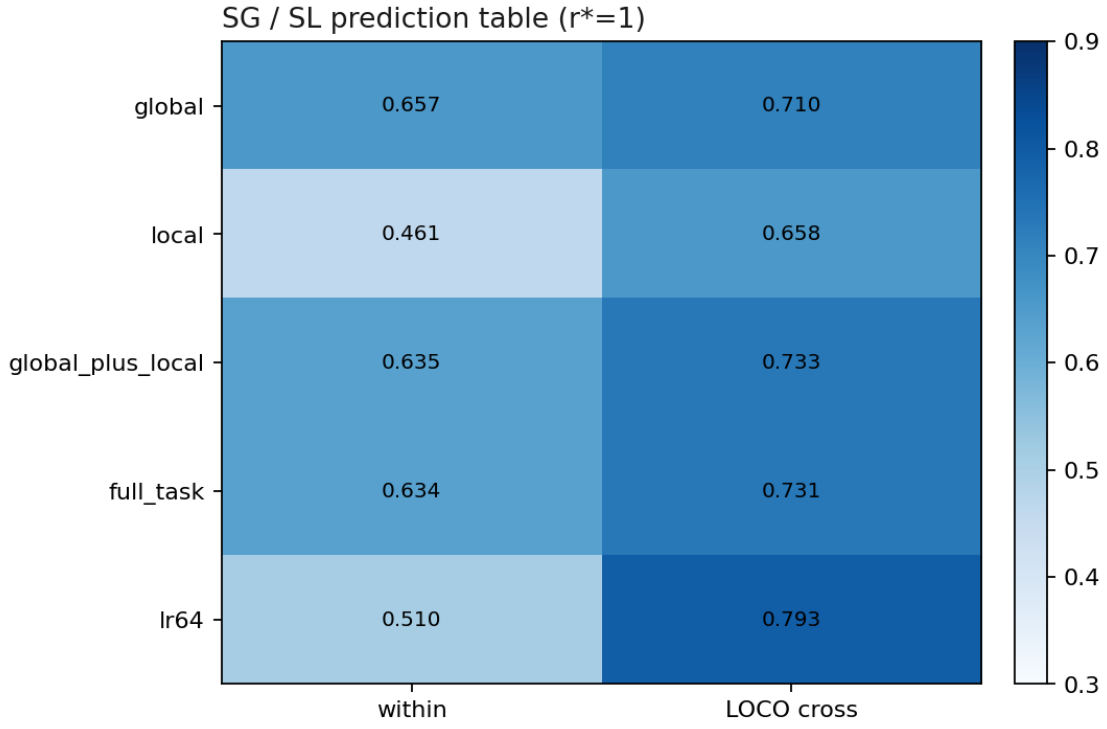
2.8 Figures

2.8.1 1. Average-projection spectrum vs nulls



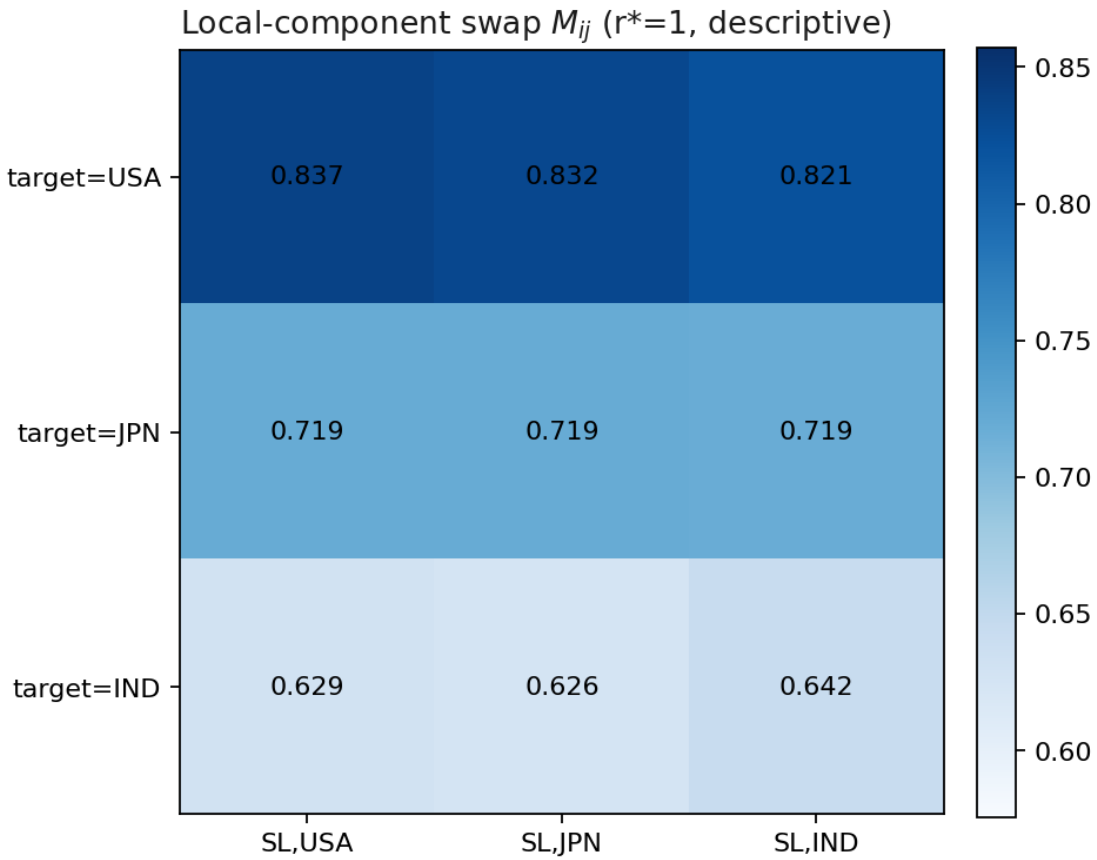
`01_avgproj_spectrum_vs_nulls.png`

2.8.2 2. SG / SL prediction table



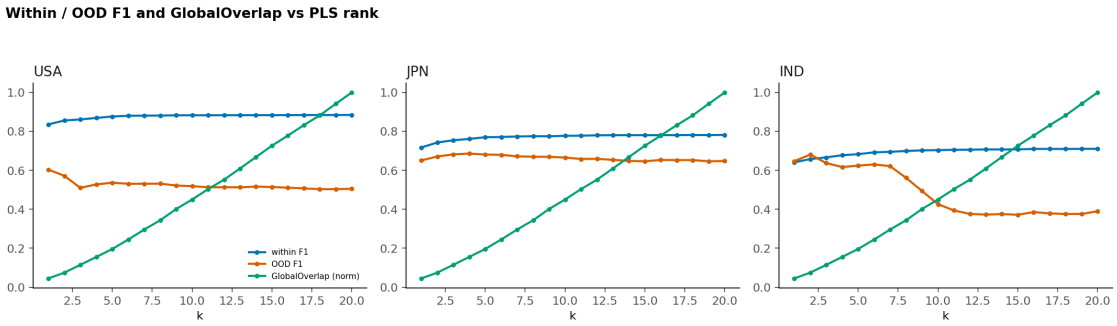
02_sg_sl_prediction_heatmap.png

2.8.3 3. Local-component swap matrix



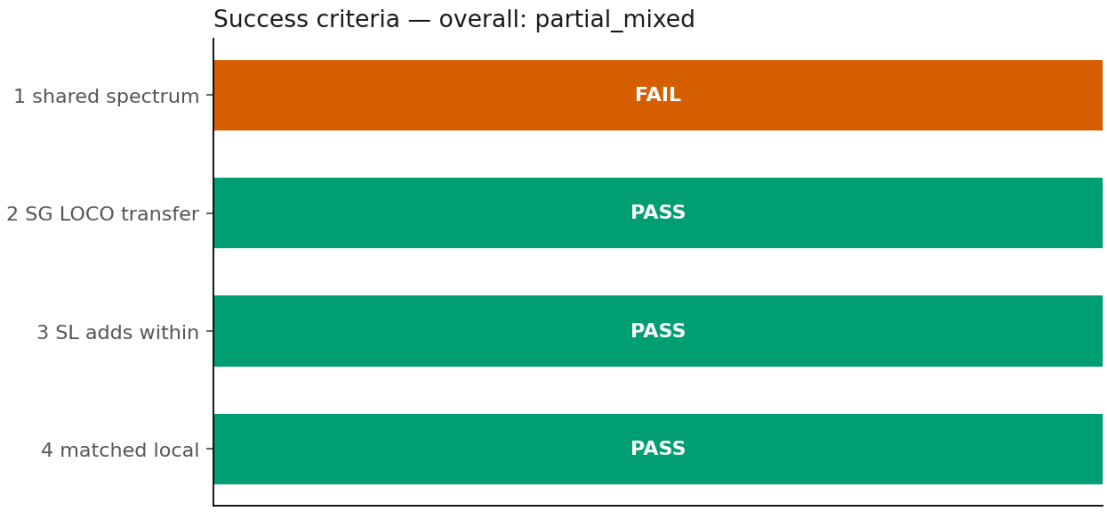
03_local_swap_matrix.png

2.8.4 4. Rank phenomenon vs GlobalOverlap



04_rank_f1_globaloverlap.png

2.8.5 5. Success-criteria dashboard



05_success_criteria_dashboard.png

[ ]:
```
from pathlib import Path
import json
import pandas as pd
from IPython.display import display, Markdown

RUN = Path(r"week5/runs/three_country_falsification_20261005_185246")
display(Markdown("### r_star"))
display(json.loads((RUN / "artifacts" / "r_star.json").read_text()))
display(Markdown("### success criteria"))
display(json.loads((RUN / "artifacts" / "success_criteria.json").read_text()))
pred = pd.read_csv(RUN / "artifacts" / "sg_sl_prediction_table.csv")
rs = json.loads((RUN / "artifacts" / "r_star.json").read_text())["r_star"]
summary = (
    pred[pred.r == rs]
    .groupby(["protocol", "mode", "representation"])["m_macro_f1"]
    .agg(["mean", "std", "count"])
    .reset_index()
)
display(Markdown("### SG/SL F1 summary at r*"))
display(summary)
display(Markdown("### Local swap diagonal advantage"))
display(pd.read_csv(RUN / "artifacts" / "local_swap_summary.csv"))
rank = pd.read_csv(RUN / "artifacts" / "rank_overlap.csv").
 ↪dropna(subset=["delta_f1_ood", "delta_global_overlap"])
if len(rank):
    corr = rank["delta_f1_ood"].corr(rank["delta_global_overlap"])
    display(Markdown(f"### corr(ΔF1_OOD, ΔGlobalOverlap) = **{corr:.3f}**"))
display(rank.head(12))
```